EXAMPLE 3 An Upper Bound

Find an upper bound for $\left|\dfrac{-1}{z^4+3z^2+2}\right|$ if $|z|=2$.

Solution By the second result in (3), the absolute value of a quotient is the quotient of the absolute values. Thus with $|-1|=1$, we want to find a positive real number M such that

$$\frac{1}{|z^4+3z^2+2|} \le M.$$

To accomplish this task we want the denominator as small as possible. Since $z^4+3z^2+2=(z^2+1)(z^2+2)$ we can use the first result in (3) and (8) to write

$$|z^4+3z^2+2| = |z^2+1|\cdot|z^2+2| \ge \big||z^2|-1\big|\cdot\big||z^2|-2\big|. \tag{12}$$

Using $|z|=2$, (12) becomes

$$\begin{aligned} |z^4+3z^2+2| \ge \big||z^2|-1\big|\cdot\big||z^2|-2\big| &= \big||z|^2-1\big|\cdot\big||z|^2-2\big| \\ &= |4-1|\cdot|4-2| \\ &= 6. \end{aligned}$$

Hence for $|z|=2$ we have

$$\left|\frac{-1}{z^4+3z^2+2}\right| = \frac{1}{|z^4+3z^2+2|} \le \frac{1}{6}.$$

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Remarks

We have seen that the triangle inequality $|z_1+z_2| \le |z_1|+|z_2|$ indicates that the length of the vector z_1+z_2 cannot exceed the sum of the lengths of the individual vectors z_1 and z_2. But the results given in (3) are interesting. The product z_1z_2 and quotient z_1/z_2, $(z_2 \neq 0)$, are complex numbers and so are vectors in the complex plane. The equalities $|z_1z_2| = |z_1|\,|z_2|$ and $|z_1/z_2| = |z_1|/|z_2|$ indicate that the lengths of the vectors z_1z_2 and z_1/z_2 are exactly equal to the product of the lengths and to the quotient of the lengths, respectively, of the individual vectors z_1 and z_2.

EXERCISES 1.2

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–4, interpret z_1 and z_2 as vectors. Graph z_1, z_2, and the indicated sum and difference as vectors.

1. $z_1 = 4+2i,\ z_2 = -2+5i;\ z_1+z_2,\ z_1-z_2$
2. $z_1 = 1-i,\ z_2 = 1+i;\ z_1+z_2,\ z_1-z_2$
3. $z_1 = 5+4i,\ z_2 = -3i;\ 3z_1+5z_2,\ z_1-2z_2$
4. $z_1 = 4-3i,\ z_2 = -2+3i;\ 2z_1+4z_2,\ z_1-z_2$
5. Given that $z_1 = 5-2i$ and $z_2 = -1-i$, find a vector z_3 in the same direction as z_1+z_2 but four times as long.

6. (**a**) Plot the points $z_1 = -2 - 8i,\ z_2 = 3i,\ z_3 = -6 - 5i$.

(**b**) The points in part (a) determine a triangle with vertices at z_1, z_2, and z_3, respectively. Express each side of the triangle as a difference of vectors.

7. In Problem 6, determine whether the points z_1, z_2, and z_3 are the vertices of a right triangle.

8. The three points $z_1 = 1 + 5i,\ z_2 = -4 - i,\ z_3 = 3 + i$ are vertices of a triangle. Find the length of the median from z_1 to the side $z_3 - z_2$.

In Problems 9–12, find the modulus of the given complex number.

9. $(1 - i)^2$

10. $i(2 - i) - 4(1 + \frac{1}{4}i)$

11. $\dfrac{2i}{3 - 4i}$

12. $\dfrac{1 - 2i}{1 + i} + \dfrac{2 - i}{1 - i}$

In Problems 13 and 14, let $z = x + iy$. Express the given quantity in terms of x and y.

13. $|z - 1 - 3i|^2$

14. $|z + 5\bar{z}|$

In Problems 15 and 16, determine which of the given two complex numbers is closest to the origin. Which is closest to $1 + i$?

15. $10 + 8i,\ 11 - 6i$

16. $\frac{1}{2} - \frac{1}{4}i,\ \frac{2}{3} + \frac{1}{6}i$

In Problems 17–26, describe the set of points z in the complex plane that satisfy the given equation. Include a plot of the set of points in the complex plane.

17. $\text{Re}((1 + i)z - 1) = 0$

18. $[\text{Im}(i\bar{z})]^2 = 2$

19. $|z - i| = |z - 1|$

20. $\bar{z} = z^{-1}$

21. $\text{Im}(z^2) = 2$

22. $\text{Re}(z^2) = \left|\sqrt{3} - i\right|$

23. $|z - 1| = 1$

24. $|z - i| = 2\,|z - 1|$

25. $|z - 2| = \text{Re}(z)$

26. $|z| = \text{Re}(z)$

In Problems 27 and 28, establish the given simultaneous inequality.

27. If $|z| = 2$, then $8 \le |\,z + 6 + 8i\,| \le 12$.

28. If $|z| = 1$, then $2 \le \left|\,z^2 - 3\,\right| \le 4$.

29. Find an upper bound for the modulus of $3z^2 + 2z + 1$ if $|z| \le 1$.

30. Find an upper and lower bound for the reciprocal of the modulus of $z^4 - 5z^2 + 6$ if $|z| = 2$. [*Hint*: $z^4 - 5z^2 + 6 = (z^2 - 3)(z^2 - 2)$.]

In Problems 31 and 32, find a number z that satisfies the given equation.

31. $|z| - z = 2 + i$

32. $|z|^2 + 1 + 12i = 6z$

Focus on Concepts

33. (**a**) Draw the pair of points $z = a + ib$ and $\bar{z} = a - ib$ in the complex plane if

$$a > 0, b > 0; a > 0, b < 0; a < 0, b > 0; \text{and} a < 0,\ b < 0.$$

(**b**) In general, how would you describe geometrically the relationship between a complex number $z = a + ib$ and its conjugate $\bar{z} = a - ib$?

(**c**) Describe geometrically the relationship between $z = a + ib$ and $z_1 = -a + ib$.

34. How would you describe geometrically the relationship between a nonzero complex number $z = a + ib$ and its

(**a**) negative, $-z$?

(**b**) inverse, z^{-1}? [*Hint*: Reread Problem 33 and then recall $z^{-1} = \bar{z}/|z|^2$.]

35. Consider the complex numbers $z_1 = 4 + i$, $z_2 = -2 + i$, $z_3 = -2 - 2i$, $z_4 = 3 - 5i$.

(**a**) Use four different sketches to plot the four pairs of points z_1, iz_1; z_2, iz_2; z_3, iz_3; and z_4, iz_4.

(**b**) In general, how would you describe geometrically the effect of multiplying a complex number $z = x + iy$ by i? By $-i$?

36. Use Definition 1.1.2 to prove that $|z| = 0$ if and only if $z = 0$.

37. Under what circumstances does $|z_1 + z_2| = |z_1| + |z_2|$?

38. Using the complex variable z, find an equation of a circle in the complex plane of radius 5 centered at $3 - 6i$. [*Hint*: Use (5).]

39. Describe the set of points z in the complex plane that satisfy $z = \cos\theta + i\sin\theta$, where θ is measured in radians from the positive x-axis.

40. Using the complex variable z, find an equation of an ellipse in the complex plane with foci $-2 + i$ and $2 + i$ whose major axis is 8 units long.

41. Suppose $z = x + iy$. In (6) of Section 1.1 we saw that x and y could be expressed in terms of z and $\bar{z}$. Use these results to express the following Cartesian equations in complex form.

(**a**) $x = 3$ (**b**) $y = 10$

(**c**) $y = x$ (**d**) $x + 2y = 8$

42. Using complex notation, find a parametric equation of the line segment between any two distinct complex numbers z_1 and z_2 in the complex plane.

43. Suppose z_1, z_2, and z_3 are three distinct points in the complex plane and k is a real number. Interpret $z_3 - z_2 = k(z_2 - z_1)$ geometrically.

44. Suppose $z_1 \neq z_2$. Interpret $\text{Re}(z_1\bar{z}_2) = 0$ geometrically in terms of vectors z_1 and z_2.

45. Suppose $w = \bar{z}/z$. Without doing any calculations, explain why $|w| = 1$.

46. Without doing any calculations, explain why the inequalities $|\text{Re}(z)| \leq |z|$ and $|\text{Im}(z)| \leq |z|$ hold for all complex numbers z.

47. Show that

(**a**) $|z| = |-z|$ (**b**) $|z| = |\bar{z}|$.

48. For any two complex numbers z_1 and z_2, show that

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2).$$

49. In this problem we will start you out in the proof of the first property $|z_1 z_2| = |z_1|\,|z_2|$ in (3). By the first result in (2) we can write $|z_1 z_2|^2 = (z_1 z_2)\overline{(z_1 z_2)}$. Now use the first property in (2) of Section 1.1 to continue the proof.

50. In this problem we guide you through an analytical proof of the triangle inequality (6).

Since $|z_1 + z_2|$ and $|z_1| + |z_2|$ are positive real numbers, we have

$|z_1 + z_2| \leq |z_1| + |z_2|$ if and only if $|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$. Thus, it suffices to show that $|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$.

(**a**) Explain why $|z_1 + z_2|^2 = |z_1|^2 + 2\text{Re}(z_1\bar{z}_2) + |z_2|^2$.

(**b**) Explain why $(|z_1| + |z_2|)^2 = |z_1|^2 + 2\,|z_1\bar{z}_2| + |z_2|^2$.

(**c**) Use parts (a) and (b) along with the results in Problem 46 to derive (6).

1.3 Polar Form of Complex Numbers

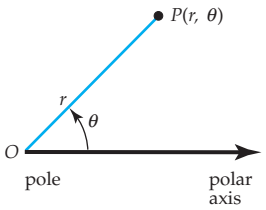


FIGURE 1.3.1 Polar coordinates

INTRODUCTION Recall from calculus that a point P in the plane whose rectangular coordinates are (x, y) can also be described in terms of **polar coordinates**. The polar coordinate system, invented by Isaac Newton, consists of point O called the **pole** and the horizontal half-line emanating from the pole called the **polar axis**. If r is a directed distance from the pole to P and θ is an angle of inclination (in radians) measured from the polar axis to the line OP, then the point can be described by the ordered pair (r, θ), called the polar coordinates of P. See FIGURE 1.3.1.

A brief review of polar coordinates can be found in Section 1.3 of the student study guide. ▶

Polar Form Suppose, as shown in FIGURE 1.3.2, that a polar coordinate system is superimposed on the complex plane with the polar axis coinciding with the positive x-axis and the pole O at the origin. Then x, y, r, and θ are related by $x = r\cos\theta$, $y = r\sin\theta$. These equations enable us to express a nonzero complex number $z = x + iy$ as $z = (r\cos\theta) + i(r\sin\theta)$ or

$$z = r(\cos\theta + i\sin\theta). \tag{1}$$

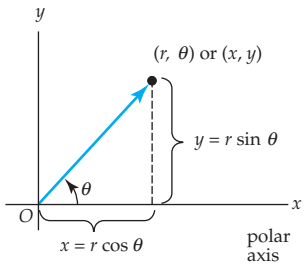


FIGURE 1.3.2 Polar coordinates in the complex plane

We say that (1) is the **polar form** or **polar representation** of the complex number z. Again, from Figure 1.3.2 we see that the coordinate r can be interpreted as the distance from the origin to the point (x, y). In other words, we shall adopt the convention that *r is never negative** so that we can take r to be the modulus of z, that is, $r = |z|$. The angle θ of inclination of the vector z, which *will always be measured in radians* from the positive real axis, is positive when measured counterclockwise and negative when measured clockwise. The angle θ is called an **argument** of z and is denoted by $\theta = \arg(z)$. An argument θ of a complex number must satisfy the equations $\cos\theta = x/r$ and $\sin\theta = y/r$. An argument of a complex number z is not unique since $\cos\theta$ and $\sin\theta$ are 2π-periodic; in other words, if θ_0 is an argument of z, then necessarily the angles $\theta_0 \pm 2\pi$, $\theta_0 \pm 4\pi$, ... are also arguments of z. In practice we use $\tan\theta = y/x$ to find θ. However, because $\tan\theta$ is π-periodic, some care must be exercised in using the last equation. A calculator will give only angles satisfying $-\pi/2 < \tan^{-1}(y/x) < \pi/2$, that is, angles in the first and fourth quadrants. We have to choose θ consistent with the quadrant in which z is located; this may require adding or subtracting π to $\tan^{-1}(y/x)$ when appropriate. The following example illustrates how this is done.

Be careful using $\tan^{-1}(y/x)$ ▶

EXAMPLE 1 A Complex Number in Polar Form

Express $-\sqrt{3} - i$ in polar form.

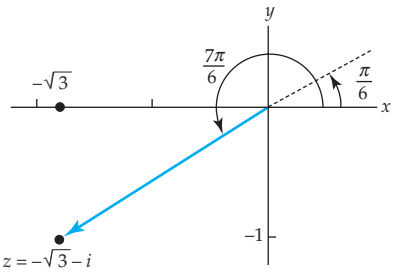


FIGURE 1.3.3 $\arg(-\sqrt{3} - i)$

Solution With $x = -\sqrt{3}$ and $y = -1$ we obtain $r = |z| = \sqrt{(-\sqrt{3})^2 + (-1)^2} = 2$. Now $y/x = -1/(-\sqrt{3}) = 1/\sqrt{3}$, and so a calculator gives $\tan^{-1}(1/\sqrt{3}) = \pi/6$, which is an angle whose terminal side is in the first quadrant. But since the point $-\sqrt{3} - i$ lies in the third quadrant, we take the solution of $\tan\theta = -1/(-\sqrt{3}) = 1/\sqrt{3}$ to be $\theta = \arg(z) = \pi/6 + \pi = 7\pi/6$. See FIGURE 1.3.3. It follows from (1) that a polar form of the number is

$$z = 2\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right). \tag{2}$$

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*In general, in the polar description (r, θ) of a point P in the Cartesian plane, we can have $r \geq 0$ or $r < 0$.